

$$2) A = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 3 & -2 \\ 1 & -2 & 3 \end{bmatrix}$$

$$R_2 = R_2 + \frac{2}{3} R_1 \downarrow$$

$$\begin{bmatrix} 3 & -2 & 1 \\ 0 & \frac{5}{3} & -\frac{4}{3} \\ 1 & -2 & 3 \end{bmatrix}$$

$$\downarrow R_3 = R_3 - \frac{1}{3} R_1$$

$$\begin{bmatrix} 3 & -2 & 1 \\ 0 & \frac{5}{3} & -\frac{4}{3} \\ 0 & -\frac{4}{3} & \frac{8}{3} \end{bmatrix}$$

$$R_3 = R_3 + \frac{4}{5} R_2 \rightarrow$$

$$\begin{bmatrix} 3 & -2 & 1 \\ 0 & \frac{5}{3} & -\frac{4}{3} \\ 0 & 0 & \frac{8}{5} \end{bmatrix}$$

Pivotlar pozitif, boyutlar pozitif (sıfır)

$$3) x_{k+1} = \frac{1}{8} (18 + y_k - z_k)$$

$$y_{k+1} = \frac{1}{5} (3 - 2x_k + 2z_k)$$

$$z_{k+1} = -\frac{1}{3} (-6 - x_k - y_k)$$

1. Iterasyon

$$x_1 = \frac{1}{8} \cdot (18) = 2,25$$

$$y_1 = \frac{1}{5} \cdot (3) = 0,6$$

$$z_1 = -\frac{1}{3} (-6) = 2$$

2. Iterasyon

$$x_2 = \frac{1}{8} (18 + 0,6 - 2) = 2,075$$

$$y_2 = \frac{1}{5} (3 - 2 \cdot 2,25 + 2 \cdot 2) = 0,5$$

$$z_2 = -\frac{1}{3} (-6 - 2,25 - 0,6) = 2,95$$

3. Iterasyon

$$x_3 = \frac{1}{8} (18 + 0,5 - 2,95) = 1,9438$$

$$y_3 = \frac{1}{5} (3 - 2 \cdot 2,075 + 2 \cdot 2,95) = 0,95$$

$$z_3 = -\frac{1}{3} (-6 - 2,075 - 0,5) = 2,8583$$

$$4) f(x) = \left(\frac{(x-4)(x-5)}{(3-4)(3-5)} \cdot 1 \right) + \left(\frac{(x-3)(x-5)}{(4-3)(4-5)} \cdot 2 \right) + \left(\frac{(x-3)(x-4)}{(5-3)(5-4)} \cdot 4 \right)$$

↓

$$f(x) = 0,5x^2 - 2,5x + 4$$

$$f(4,5) = 2,875$$

5) Bölünmüş Farklar Tablosu

x	y	1. Farklar	2. Farklar	3. Farklar
1	0			
		$\frac{-2 - 0}{2 - 1} = -2$		
2	-2		$\frac{2 - (-2)}{3 - 1} = 2$	
		$\frac{0 - (-2)}{3 - 2} = 2$		$\frac{0.5 - 2}{4 - 1} = -0.5$
3	0		$\frac{3 - 2}{4 - 2} = 0.5$	
		$\frac{3 - 0}{4 - 3} = 3$		
4	3			

$$f(x) = 0 + ((x-1) \cdot -2) + (x-1)(x-2) \cdot 2 + (x-1)(x-2)(x-3) \cdot 0.5$$

$$\downarrow$$

$$f(x) = -0.5x^3 + 5x^2 - 13.5x + 9$$

$$f(2.3) = -1.68$$

$$y = mx + b$$

$$6) \sum x = 1 + 2 + 3 + 4 + 5 + 6 + 7 = 28$$

$$\sum y = 0,5 + 2,5 + 2 + 4 + 3,5 + 6 + 5,5 = 24$$

$$\sum x \cdot y = 0,5 + 5 + 6 + 16 + 17,5 + 36 + 38,5 = 89,5$$

$$\sum x^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2 = 140$$

$$n = 7$$

$$m = \frac{(7 \cdot 89,5 - 28 \cdot 24)}{(7 \cdot 140 - 28^2)} = 0,785714$$

$$b = \frac{(24 - 0,785714 \cdot 28)}{7} = -0,357143$$

$$y = 0,785714x - 0,357143$$